• **SELECT SHIFT TRANSMISSION (SST) OPERATION—VEHICLES WITH 5R55N TRANSMISSION—UNDERSTANDING RANGE SELECTOR OPERATION—SERVICE TIP** • **TRANSMISSION—5R55N—UNDERSTANDING SELECT SHIFT TRANSMISSION (SST) RANGE SELECTOR OPERATION—SERVICE TIP**	**Article No. 99-21-4**

LINCOLN: 2000 LS

ISSUE

This TSB article is being made available to use as an aid in discussing with the customer how to properly utilize the SST range selector.

ACTION

The following information provides a better understanding on how the SST range selector controls 5R55N transmission operation in the Lincoln LS.

Range Selection Using The SST Selector
The optional Select Shift Transmission (SST) version of the transmission range selector has seven positions. This version also allows the driver to manually select the appropriate upshift or downshift and gear range. The positions of this version are: P, R, N, D5 or D4 and + or - (Figure 1).

Park
In the PARK position:

- There is no powerflow through the transmission
- The parking pawl locks the output shaft to the case
- The engine may be started
- The ignition key may be removed

Reverse
In the REVERSE position:

- Vehicle may be operated in a rearward direction, at a reduced gear ratio
- Engine braking will occur
- Backup lamps are illuminated

Neutral
In the NEUTRAL position:

- There is no powerflow through the transmission
- The output shaft is not held and is free to turn
- The engine may be started

D5 (Overdrive) Position
D5 (overdrive) is the normal position for most forward driving. The D5 (overdrive) position provides:

- Automatic shift 1-5 and 5-1
- Apply and release of the torque converter clutch
- Maximum fuel economy during normal operation
- Engine braking with the coast clutch applied in 5th gear

D4 (Overdrive Canceled) Position
The D4 (overdrive canceled) position provides:

- Automatic shifts 1-4 and 4-1
- Apply and release of the torque converter clutch
- Engine braking in 4th gear

+ and - Position Operation
These positions allow the driver manually select the appropriate upshift (+) or downshift (-) and gear

range.

- Can only be entered from the D5 position
- Gear ranges 1-5 provide the same function and ratio as found in the D5 or D4 automatic mode positions
- Transmission will not upshift or downshift unless the selector lever is moved forward or rearward
- One tap forward (+) will allow the transmission to upshift one gear range
- One tap rearward (-) will allow the transmission to downshift one gear range
- Upshifts are allowed at any vehicle speed
- 2nd gear start up - normally. A 3rd gear start up will be allowed only if selected. All other selections will revert to 2nd gear.
- If the driver starts in 2nd, and does not pass 60% throttle, the transmission will remain in 2nd gear. No automatic shift will occur.
- If the driver passes 60% throttle, then a 2-1 automatic kickdown will occur, scheduled by the Powertrain Control Module (PCM) as used in D5. Once this has occurred, first gear will hold until 2nd gear is selected by the driver moving the shift handle. No automatic upshift is possible under this condition.
- Once a kickdown has occurred, manual shifting in and out of 1st gear is allowed until the next time the vehicle stops
- A shift to 1st with the vehicle at rest is never allowed
- When downshifting at normal road speeds, the transmission will only allow a downshift into next lower gear
- If shifter (-) is tapped more than once in rapid succession, the transmission will downshift only into the next gear, then when the vehicle reaches a speed below a calibrated entry speed, the transmission will allow a downshift into the next lower gear if again selected by the driver. This is to prevent engine and transmission damage and to keep the engine and transmission within allowable rpm ranges.
- An electronic indicator on the instrument cluster will display the selected gear

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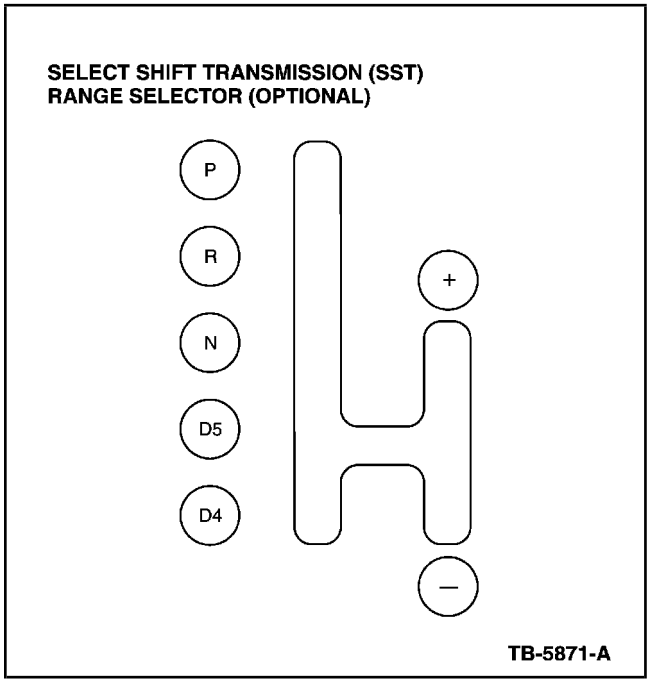

Figure 1 - Article 99-21-4